Determination of Drought Triggering Probability via Bayesian Network

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Abstract

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1 Introduction

Drought Anticipatory Action (DAA)

bnet (Bayesian network) (see Ref.[3])

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

2 Notation and Conventions

We will use $\mathbb{1}(S)$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if statement S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this article, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x.

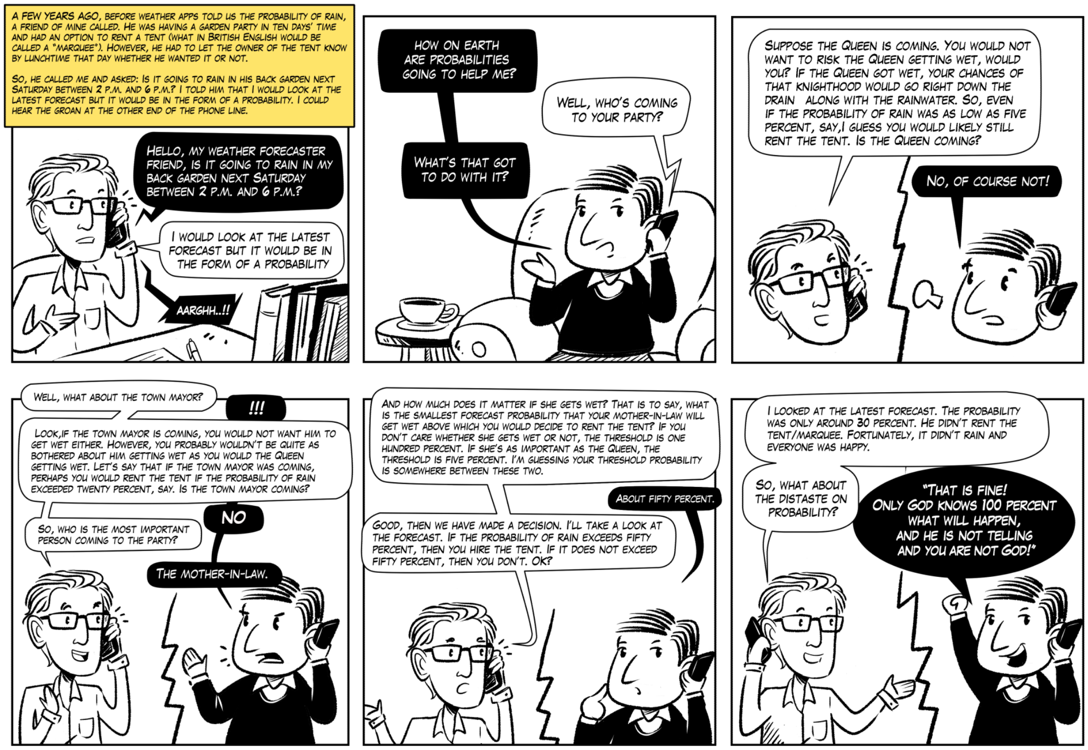
Let

$N =$ the number of times (seasons) in which predicted and observed droughts were compared.

$N_{x|a} =$ number of events in which a number of $a \in \{0, 1\}$ droughts occurred and a number of $x \in \{0, 1\}$ droughts were predicted. For example, $N_{1|0}$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm).

$N_{0|0} =$ **True Negatives** (TN), correct rejection

$N_{1|1} =$ **True Positives** (TP), hit



Source: Palmer, Tim. The Primacy of Doubt: From climate change to quantum physics, how the science of uncertainty can help predict and understand our chaotic world. Oxford University Press, 2022.

Figure 1: Illustration from Tim Palmer's book Ref.[2]

$N_{0|1}$ = **False Negatives** (FN), miss, type II error or underestimation
$N_{1|0}$ = **False Positives** (FP), false alarm, type I error or overestimation

$$N_{|a} = \sum_{x'} N_{x'|a}, \quad N_{x|} = \sum_{a'} N_{x|a'} \tag{1}$$

$$N = \sum_x \sum_a N_{x|a} = \sum_a N_{|a} = \sum_x N_{x|a} \tag{2}$$

$$P(x|a) = R_{x|a} = \frac{N_{x|a}}{N_{|a}},\ P'(a|x) = R'_{a|x} = \frac{N_{x|a}}{N_{x|}} \tag{3}$$

$$\sum_x P(x|a) = \sum_a P'(a|x) = 1 \tag{4}$$

Note that both $P(x|a)$ and $P'(a|x)$ are conditional probability distributions.

Note that $N_{x|a}$ for $x, a \in \{0, 1\}$ yields 4 degrees of freedom (dofs). Instead of those 4 dofs, we will use N and 3 others dofs, namely, hit rate (HR), false alarm rate (FAR) and area under ROC curve (AUROC). HR, FAR and AUROC are defined in Appendix A

3 DAA bnet

Let

$\kappa = (\sigma, \delta)$ = the drought kind, where $\sigma \in \{\text{MAM, JJAS}\}$[1] is the season, and $\delta \in \{\text{Moderate, Extreme, Severe}\}$ is the drought degree.

I_κ = **Impact** (i.e., **severity**) of drought is defined as minus SPI (higher rain, higher SPI, less severe drought)

$P(I_\kappa)$ = likelihood, probability that a drought of severity I_κ will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for $ECMWF$ (European Centre for Medium-Range Weather Forecasts)

$$Risk_\kappa = E[\underline{I}_\kappa] = \frac{1}{M}\sum_{\mu=1}^{M} I_{\kappa,\mu} \tag{5}$$

$$= \sum_{I_\kappa} P(I_\kappa) I_\kappa \tag{6}$$

A **risk matrix** is a 4×4 matrix with X axis equal to impact (I) and Y axis equal to likelihood P, with I binned into four values and P binned into 4 values. The matrix elements give the value of some arbitrary function $f(I, P)$ defined as risk. Usually, $f(I, P) = IP$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[5]) agrees.

$\sigma \backslash \delta$	Moderate	Extreme	Severe
MAM	-0.55	-0.98	-0.99
JJAS	-0.41	-0.98	-0.99

Table 1: SPI threshold for Karamoja, Uganda. Cells give $-I^*_\kappa$, where $\kappa = (\sigma, \delta)$

I^*_κ = impact threshold for drought (See Table 1 for examples of I^*_κ.)

$$P(\underline{I}_\kappa \geq I^*_\kappa) = \frac{1}{M}\sum_{\mu=1}^{M} \mathbb{1}(I_{\kappa,\mu} \geq I^*_\kappa) \tag{7}$$

Let $X^i_\kappa \in \{0, 1\}$ for $i = 1, 2, 3, \ldots, nx$ represent vulnerability to local problems such as high road density, high population density, historical catastrophic events, etc. It equals 1 if a particular danger is present, and 0 if not.

Let

$$N_\kappa = \sum_{i=1}^{nx} X^i_\kappa \tag{8}$$

[1]Letters here stand for the first letter of the months of the year.

$\epsilon \in \mathbb{R}$ be a normally distributed random variable with mean zero and some a priori given deviation.

RC^i_κ = replacement cost for vulnerability X^i_κ

IC^i_κ = insurance (i.e., preventive) cost for vulnerability X^i_κ

Note that insurance always costs more than replacement, so

$$0 \leq IC^i_\kappa \leq RC^i_\kappa \tag{9}$$

$P^i_{danger,\kappa} = \frac{IC^i_\kappa}{RC^i_\kappa}$ probability of danger for Vulnerability X^i_κ.

Note that less danger means less insurance cost.

P^*_κ = probability threshold for drought

$$P^*_\kappa = f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) + \epsilon \tag{10}$$

where we model the function $f_\kappa()$ by

$$f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) = \begin{cases} 1 - \frac{1}{N_\kappa}\sum_{i=1}^{nx} P^i_{danger,\kappa} X^i_\kappa & \text{if } N_\kappa \neq 0 \\ 1 & \text{if } N_\kappa = 0 \end{cases} \tag{11}$$

T_κ = trigger for drought

P^*_κ = threshold probability for drought warning

$$T_\kappa = \mathbb{1}\left[P(\underline{I}_\kappa \geq I^*_\kappa) > P^*_\kappa\right] \tag{12}$$

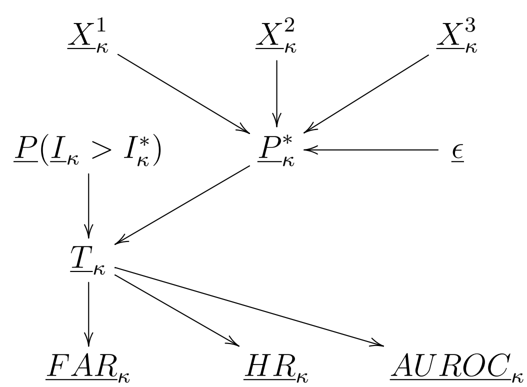


Figure 2: DAA bnet. We use one of these bnets for each κ value and for each grid cell of 0.25 degrees in latitude by 0.25 degrees in longitude. 1 degree in latitude or longitude is approximately 100 km.

The TPMs[2] printed in blue, for the bnet Fig.2, are as follows.[3]

[2]TPM=Transition Probability Matrix. This is the same thing as CPT=Conditional Probability Table

[3]A deterministic node satisfying the structural equation $y = f(x_1, x_2)$, has the TPM: $P(y|x_1, x_2) = \delta(y, f(x_1, x_2))$, where $\delta(a, b) = \mathbb{1}(a = b)$ is the Kronecker delta function.

$$P^*_\kappa = f_\kappa(X^1_\kappa, X^2_\kappa, \ldots, X^{nx}_\kappa) + \epsilon \tag{13}$$

$$T_\kappa = \mathbb{1}[P(\underline{I}_\kappa > I^*_\kappa) > P^*_\kappa] \tag{14}$$

$$P(Y^i_\kappa | T_\kappa) = \text{to be learned empirically from dataset} \tag{15}$$

where $(Y^1_\kappa, Y^2_\kappa, Y^3_\kappa) = (FAR_\kappa, HR_\kappa, AUROC_\kappa)$.

Algorithm for finding best triggers T^*_κ:

Enter bnet Fig.2 into bnet software PyAgrum (Ref.[1]). Enter into software available evidence for nodes $P(\underline{I}_\kappa > I^*_\kappa)$, (X^1, X^2, X^3) and $(Y^1_\kappa, Y^2_\kappa, Y^3_\kappa)$. Allow software to infer the distribution of $\underline{T}^*_\kappa \in \{0, 1\}$. Choose for T^*_κ the state with maximum probability.

A Appendix: Functions of $N_{x|a}$

Terminology Table Adapted from Wikipedia Ref.[4]

positive events (P): $P = N_{|1} = \sum_x N_{x|1}$, the number of real positive events in the data

negative events (N): $N = N_{|0} = \sum_x N_{x|0}$, the number of real negative events in the data

sensitivity, recall, true positive rate (TPR), or hit rate (HR):

$$TPR = R_{1|1} = \frac{N_{1|1}}{N_{|1}} = \frac{N_{1|1}}{N_{1|1} + N_{0|1}} = 1 - R_{0|1} \tag{16}$$

specificity, selectivity or true negative rate (TNR):

$$TNR = R_{0|0} = \frac{N_{0|0}}{N_{|0}} = \frac{N_{0|0}}{N_{0|0} + N_{1|0}} = 1 - R_{1|0} \tag{17}$$

precision or positive predictive value (PPV):

$$PPV = \frac{N_{1|1}}{N_{1|1} + N_{1|0}} = 1 - FDR \tag{18}$$

negative predictive value (NPV):

$$NPV = \frac{N_{0|0}}{N_{0|0} + N_{0|1}} = 1 - FOR \tag{19}$$

miss rate or false negative rate (FNR):

$$FNR = R_{0|1} = \frac{N_{0|1}}{N_{|1}} = \frac{N_{0|1}}{N_{0|1} + N_{1|1}} = 1 - R_{1|1} \tag{20}$$

fall-out, false positive rate (FPR), or false alarm rate (FAR):

$$FPR = R_{1|0} = \frac{N_{1|0}}{N_{|0}} = \frac{N_{1|0}}{N_{1|0} + N_{0|0}} = 1 - R_{0|0} \tag{21}$$

false discovery rate (FDR):

$$FDR = \frac{N_{1|0}}{N_{1|0} + N_{1|1}} = 1 - PPV \tag{22}$$

false omission rate (FOR):

$$FOR = \frac{N_{0|1}}{N_{0|1} + N_{0|0}} = 1 - NPV \tag{23}$$

accuracy (ACC):

$$ACC = \frac{N_{1|1} + N_{0|0}}{N_{|1} + N_{|0}} = \frac{N_{1|1} + N_{0|0}}{N_{1|1} + N_{0|0} + N_{1|0} + N_{0|1}} \tag{24}$$

balanced accuracy (BA):

$$BA = \frac{R_{1|1} + R_{0|0}}{2} \tag{25}$$

F1 score is the harmonic mean of precision and sensitivity:

$$F_1 = 2 \times \frac{PPV \times R_{1|1}}{PPV + R_{1|1}} = \frac{2N_{1|1}}{2N_{1|1} + N_{1|0} + N_{0|1}} \tag{26}$$

The area under the ROC (Receiver Operating Characteristic) curve, known as AUC or AUROC, is defined by Fig.3. One set of 4 values $N_{x|a}$ where $a, x \in \{0, 1\}$, only gives one point of the ROC curve. To get a full ROC curve, we need to vary a parameter. In the case of DAA, we vary geographical location, for example.

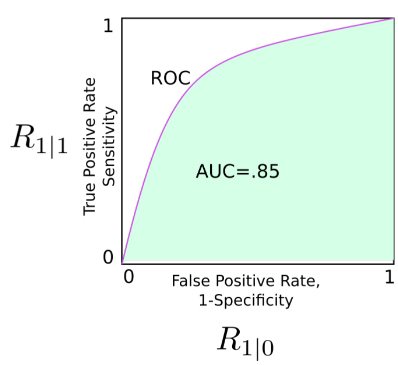


Figure 3: Example of ROC. Green shaded area is the AUC of the ROC.

References

[1] Agrumery. PyArum, free open source app at github. https://github.com/agrumery/aGrUM.

[2] Tim N. Palmer. *The Primacy of Doubt: From climate change to quantum physics, how the science of uncertainty can help predict and understand our chaotic world*. Oxford University Press, 2022.

[3] Robert R. Tucci. Bayesuvius (free book). https://github.com/rrtucci/Bayesuvius.

[4] Wikipedia. Receiver operating characteristic. https://en.wikipedia.org/wiki/Receiver_operating_characteristic.

[5] Wikipedia. Risk matrix. https://en.wikipedia.org/wiki/Risk_matrix.